

Optimization

Minimization and Karnaugh Maps

- Simplify (or minimize) POS functions (circuits):

Example 1: Find the minimum cost POS expression for the following using a K-map

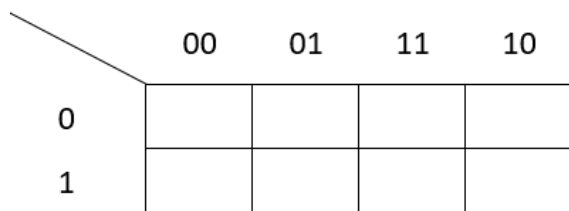
$$f(x_1, x_2, x_3) = \overline{x_1}\overline{x_2}x_3 + x_1\overline{x_2}x_3 + x_1\overline{x_3}$$

x_1	x_2	x_3
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

	00	01	11	10
0				
1				

Example 2: Give the minimum cost POS representation of the following logical expression

$$f(x_1, x_2, x_3) = \prod M(0, 1, 4, 5, 6)$$



	00	01	11	10
0				
1				

Example 3: Find the minimum cost POS expression for the following function.

$$f(x_1, x_2, x_3, x_4) = \sum m(2, 3, 9, 10, 11, 13)$$

	00	01	11	10
00				
01				
11				
10				

Don't Cares in K-Maps

- Any function may have input values that will never occur => the output value is considered as a “don't care”, X or d
- Don't cares in the K-map may reduce the cost of the logic circuit

Given $f(x_1, \dots, x_4) = \sum m(2, 4, 5, 6, 10) + D(12, 13, 14, 15)$, find the minimum cost SOP and POS expressions.

$x_1 x_2$ $x_3 x_4$	00	01	11	10
00	0 ⁰	1 ⁴	d ¹²	0 ⁸
01	0 ¹	1 ⁵	d ¹³	0 ⁹
11	0 ³	0 ⁷	d ¹⁵	0 ¹¹
10	1 ²	1 ⁶	d ¹⁴	1 ¹⁰

$x_1 x_2$ $x_3 x_4$	00	01	11	10
00				
01				
11				
10				

Example: Find the minimum cost SOP and POS functions.

	00	01	11	10
00	0	1	d	0
01	0	1	d	0
11	1	0	d	1
10	1	1	d	1

	00	01	11	10
00	0	1	d	0
01	0	1	d	0
11	1	0	d	1
10	1	1	d	1

Terminology

Literal: Each appearance of a variable, either uncomplemented or complemented, is called a literal.

Implicant: A product term for which the function is equal to 1.

Prime Implicant: An implicant that cannot be combined with another to reduce number of literals and still be an implicant.

Essential Prime Implicant: If a prime implicant includes a minterm for which $f = 1$ that is not included in any other prime implicant, is called essential prime implicant.

Cover: A collection of implicants that covers all the minterms of the function. A cover defines a particular implementation of the function.

Example:

Given the following function:

$$\text{Four-variable function } f(x_1, \dots, x_4) = \sum m(2, 3, 5, 6, 7, 10, 11, 13, 14).$$

1. How many possible implicants does the given function f have?
2. Find all prime implicants.
3. Find all essential prime implicants.
4. Derive the minimum cost cover.

$x_1 x_2$ $x_3 x_4$	00	01	11	10
00				
01				
11				
10				